

Modeling Rural Healthcare Accessibility Using Road Network Analysis in Python

GSP 318 - Cal Poly Humboldt

May 12th, 2026

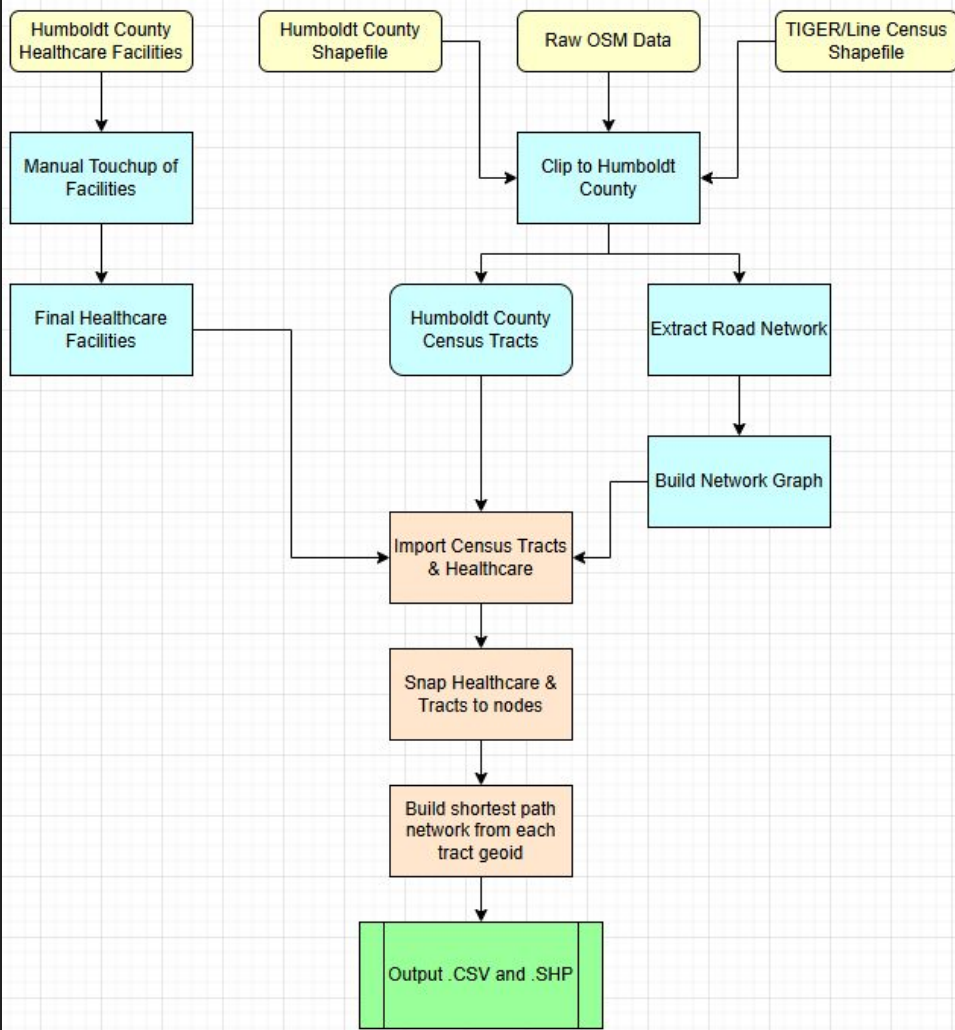
Julian Nelson

Project Goals

- Create an automated healthcare accessibility workflow for County level analysis
- Develop manual and API based scripts
- Output travel time datasets to be mapped
- Identify areas with long travel times (low accessibility) to healthcare services

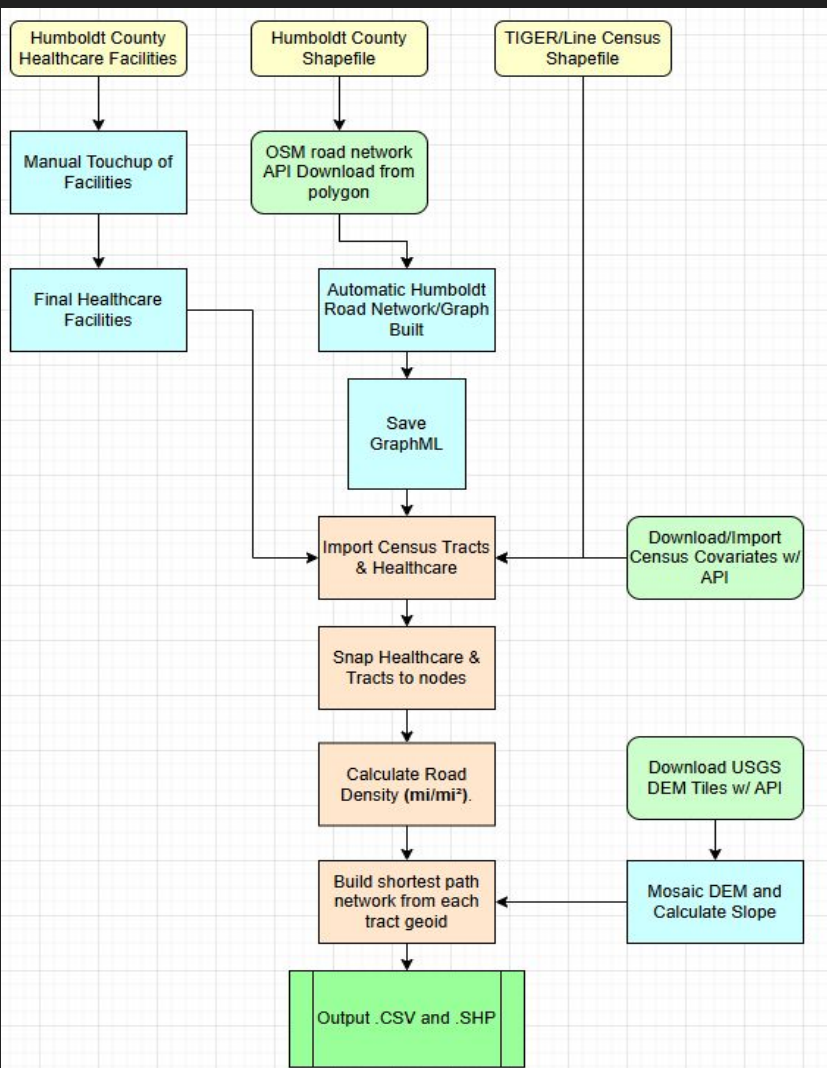
Software & Libraries

- **Python**
 - **OSMnx** - Open Street Maps Python Library
 - **NetworkX** - Library for performing Graph/Network functions
 - **GeoPandas** - For various geospatial functions
 - **Rasterio** - For DEM handling
- **ArcPro** - Mapping Final outputs
- **RStudio** - Statistical Analysis of model/covariates
 - **Mgcv** - Library for creating Generalized Additive Models



Manual Download

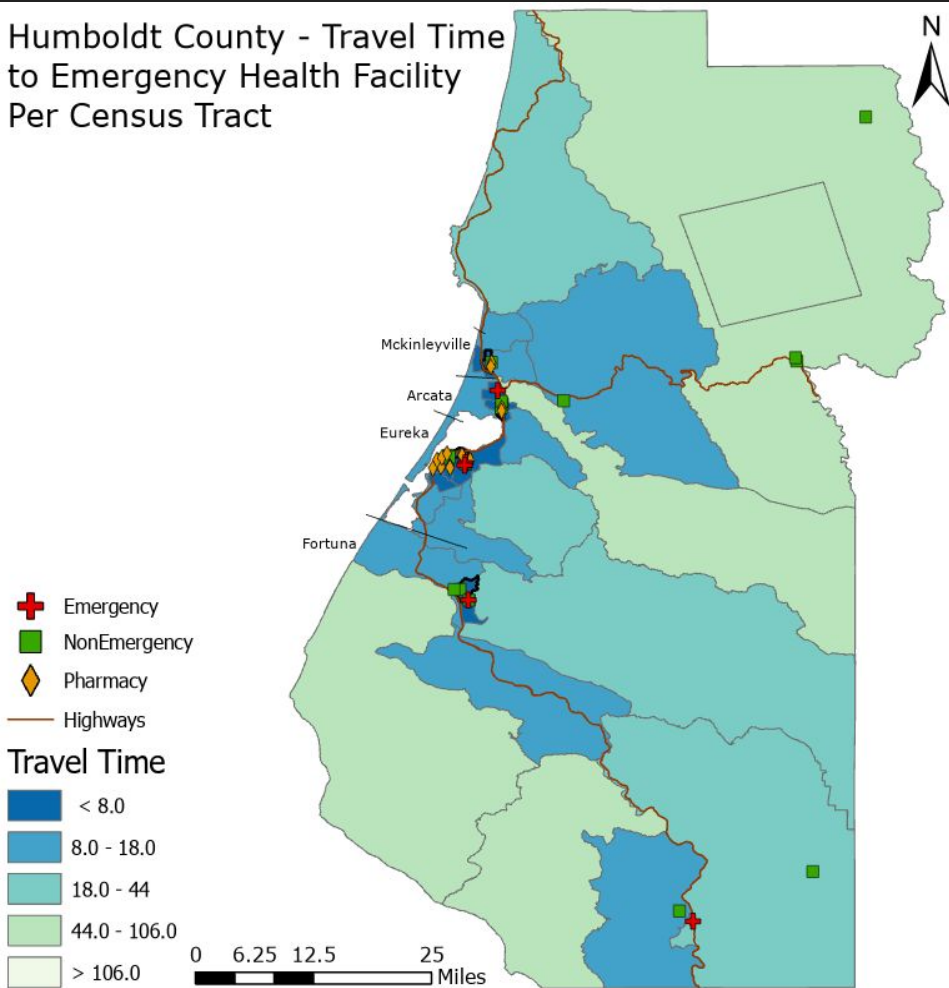
- Clipping in ArcPro
- Manual removal of outdated healthcare facilities
- Large OSM files - 2 Million+ plus road segments in NorCal
- Disconnected road segments
- Time consuming
- Harder Reproducibility



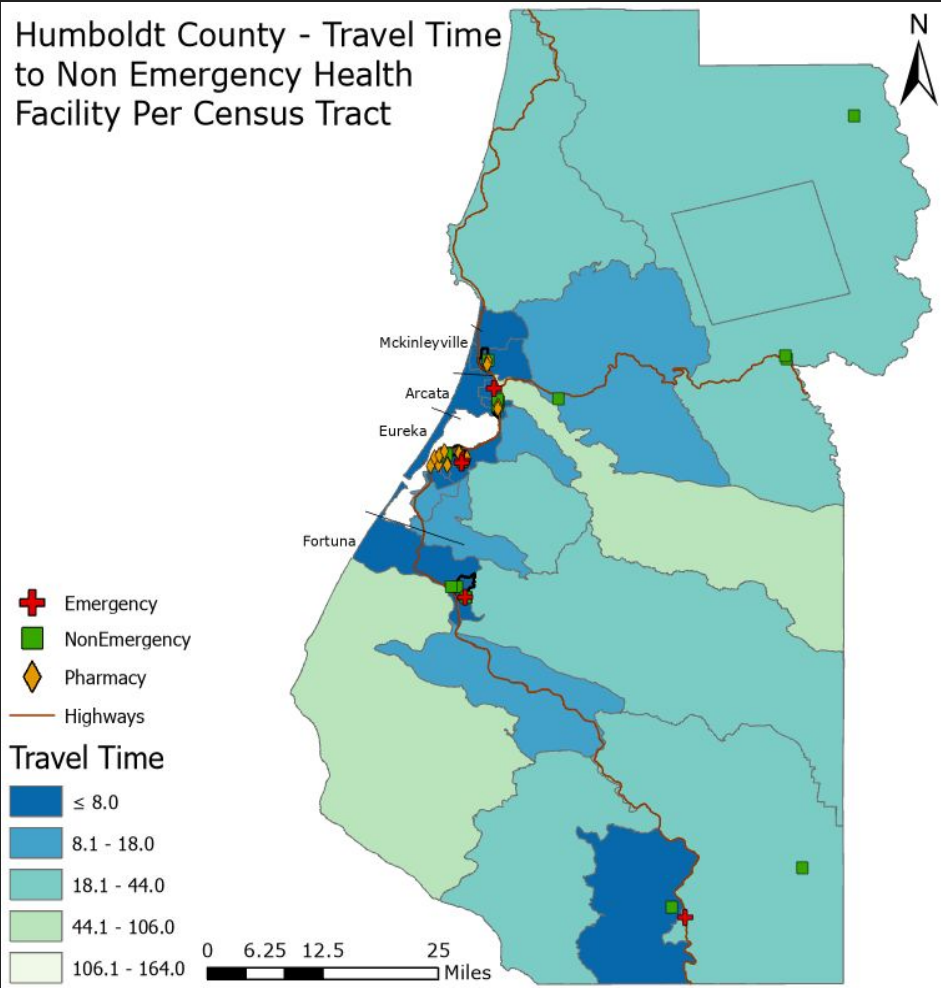
API Download

- More automated
- Reproducible with similar inputs
- Smaller OSM files already clipped
- Better routing accuracy, less disconnected segments
- Much Faster
- Simpler Script

Humboldt County - Travel Time
to Emergency Health Facility
Per Census Tract



Humboldt County - Travel Time
to Non Emergency Health
Facility Per Census Tract



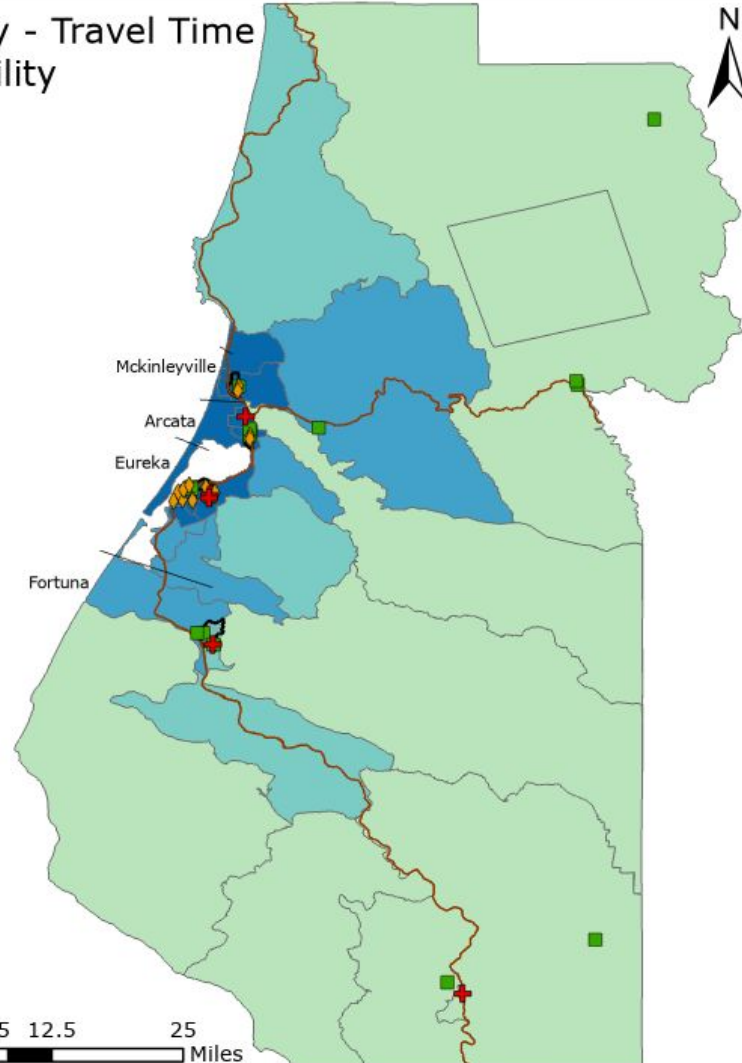
Humboldt County - Travel Time to Pharmacy Facility Per Census Tract

- Emergency
- NonEmergency
- Pharmacy
- Highways

Travel Time

- ≤ 8.0
- 8.1 - 18.0
- 18.1 - 44.0
- 44.1 - 106.0
- 106.1 - 164.0

0 6.25 12.5 25 Miles



Python Library Documentation

- OSMnx: <https://osmnx.readthedocs.io/en/stable/>
- NetworkX: <https://networkx.org/documentation/stable/reference/index.html>
- Pandas: https://pandas.pydata.org/docs/user_guide/index.html#user-guide
- GeoPandas: <https://geopandas.org/en/stable/docs.html>
- Matplotlib: <https://matplotlib.org/stable/index.html>
- Rasterio: <https://rasterio.readthedocs.io/en/stable/intro.html>

References

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- U.S. Census Bureau. (2024). TIGER/Line Shapefiles: Census Tracts, California. Retrieved from <https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html>
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